Paleomagnetism of the Cumberland Group and other Paleozoic formations[1]

J. L. Roy
Geomagnetic Division, Dominion Observatory, Ottawa, Canada

Received April 8, 1969
Accepted for publication May 8, 1969

Thermal demagnetization of red bed samples from the Cumberland Group (Pennsylvanian) of Nova Scotia indicates that the fossil magnetization is composed mainly of a thermally distributed (up to 650 °C) magnetization. The direction changes between heatings are small, indicating that this magnetization was acquired in a field of constant polarity. A fold test (10 sites) shows that the magnetization predates the folding, which according to geological evidence occurred shortly after deposition. The results (direction 172°, +16°; $k = 85$; pole 36° N, 125° E) are compared with other results obtained from Paleozoic formations. Comparison between continents indicate that a 30° apparent polar change during the Paleozoic is probable. From the Dominion Observatory results, it is suggested that at least half of the change took place during late Paleozoic times and probably during the Pennsylvanian Period.

Une étude par recuits successifs d'échantillons sédimentaires du groupe de Cumberland (Pennsylvanien) en Nouvelle Ecosse révèle que l'aimantation fossile est composée principalement d'une aimantation dont la gamme de températures de blocage s'étend jusqu'à au moins 650 °C. La direction d'aimantation, cependant, subit très peu de changement et l'on en déduit que cette aimantation fut acquise dans un champ de polarité unique. Un test de repli (10 localités) démontre que son acquisition eut lieu avant un plissement du terrain qui, selon l'évidence géologique, s'effectua peu de temps après déposition. Les résultats (direction 172°, 16°; $k = 85$; pôle 36° N, 125° E) peuvent nous indiquer, dès lors, la direction du champ magnétique et la position du pôle, par rapport à la Nouvelle-Ecosse, du Pennsylvanien moyen. Cette position cadre bien avec l'ensemble des positions données par d'autres formations du Paléozoique. Une comparaison de résultats provenant de différents continents démontre qu'un déplacement apparent de 30° du pôle pendant le Paléozoique est concevable. D'après les données obtenues à l'Observatoire fédéral, il semblerait qu'au moins la moitié de ce déplacement eut lieu pendant le Paléozoique supérieur et peut-être plus précisement pendant le Pennsylvanien.

Introduction

One project of the Dominion Observatory has been to obtain a record of the Upper Paleozoic geomagnetic field, from paleomagnetic studies of sedimentary rock formations of Eastern Canada. In this paper, results from red sediments of the Cumberland Group in Nova Scotia are described. These and previous results of this project are then summarized, and from this summary certain broad features of the geomagnetic field in Eastern Canada are apparent.

Geology and Sampling

The Pennsylvanian of Nova Scotia is divided, from the bottom up, into three groups: the Riversdale, the Cumberland, and the Pictou. The Cumberland is well exposed in the northeastern part of the province. At Joggins, 7 sites (inset of Fig. 1) spanning a stratigraphic thickness of 725 m were sampled; the attitude of the beds varies slightly, averaging 25° at 182° true bearing. The rocks sampled were fine- to medium-grained sandstones ranging from purple (sites 1–3) to deep red (sites 4–6) to pink (site 7) in color. At site 8 on Chignecto Bay, the samples were medium-grained red sandstones dipping 22° at 160°, and at the remaining sites (all inland) they were deep red (site 9) and purple (sites 10 and 11) sandstones dipping 29° at 344°, 15° at 011°, and 17° at 028° respectively. The variation in dip from site to site allows a fold test to be made. According to Bell (1944), there is widespread disconformity and local unconformity between the Pictou and Cumberland strata. In the description on the Oxford sheet (geological map 410A, Canada Department of Mines and Resources, 1938), it is stated: "The Pictou rest unconformably on the Cumberland strata". East of Oxford, where Pictou beds overlie Cumberland strata, the dips of the Pictou beds are $\simeq 10°$ shallower than those of the Cumberland. It is reasonable therefore to conclude that part at least of the folding of the Cumberland beds occurred before Pictou time.

[1]Contributions from the Dominion Observatory, No. 274.

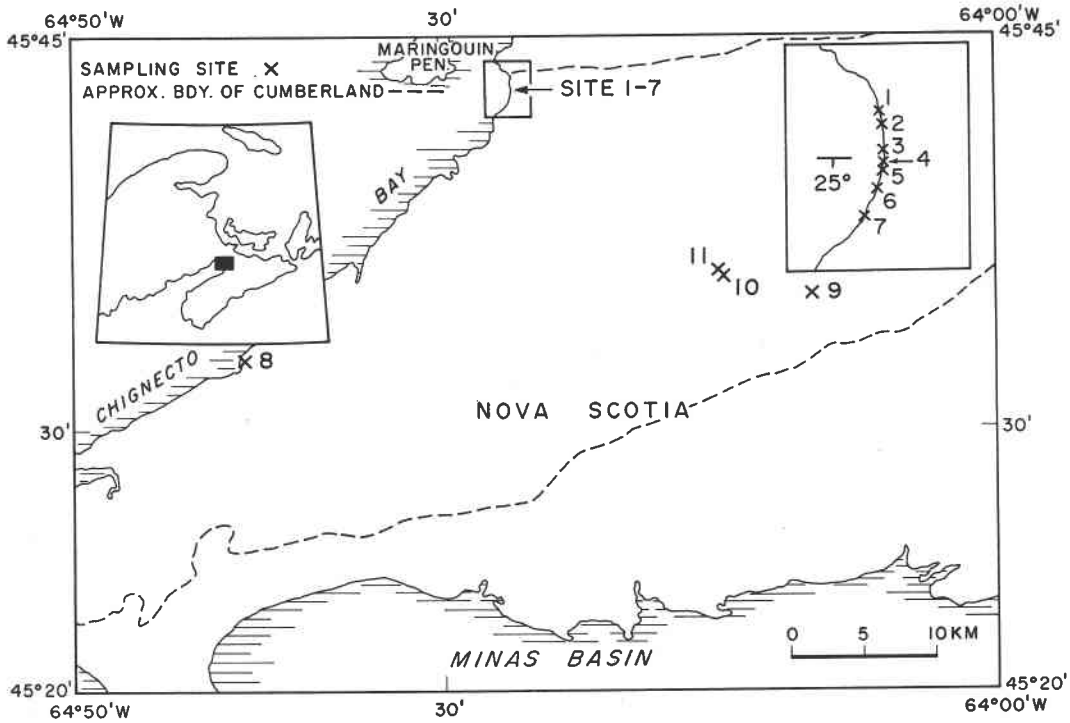


FIG. 1. Locality map showing the sampling sites.

Two to nine samples were taken from each of the 11 sites as shown in Fig. 1 and listed in Table I. Most samples were oriented drilled cores. A few were oriented blocks. One to three cylindrical specimens (2.5 cm diameter and 2.2 cm height) were cut from each core or block. The directions and intensities were measured on the Dominion Observatory astatic magnetometer. Sample means were used as the basic data in the computations. In this paper M_N and M_T refer to data before and after thermal demagnetization at temperature T.

TABLE I

Analysis of intensities and directions before (M_N) and after (M_T) thermal cleaning at temperature T

			M_N					M_T						Pole position Lat (°N),
Site	N_{sp}	N_{sa}	M_1	D°, I°	R	k	α_{95}	T	M_1	D°, I°	R	k	α_{95}	Long (°E)
1	10	4	14.4	172, +16	3.934	45	14	550	11.3	169, +14	3.937	48	13	36, 129
2	6	4	21.0	172, +30	3.984	186	07	550	8.7	171, +22	3.951	62	12	32, 126
3	10	6	26.1	178, +18	5.983	300	04	550	14.5	181, +14	5.977	225	04	37, 115
4	8	5	26.0	172, +14	4.980	199	05	550	9.2	174, +09	4.981	212	05	40, 123
5	13	7	21.3	166, +36	6.688	20	14	550	9.3	173, +27	6.891	55	08	30, 124
6	12	6	21.1	174, +21	5.950	100	07	635	9.6	177, +16	5.932	73	08	36, 120
7	*14*	*9*§	*3.4*	*178, +39*	*8.426*	*14*	*15*	*550*	*2.4*	*165, +09*	*2.991*§	—	—	*38, 134*
8	6	6	6.0	152, +04	5.876	40	11	550	4.4	163, +05	5.916	60	09	40, 137
9	2	2	3.6	155, +55	1.991	—	—	550	1.9	166, +26	2.000	—	—	29, 131
10	6	4	10.2	179, +24	3.953	64	12	550	7.0	178, +14	3.972	108	09	37, 118
11	9	4	18.2	170, +16	3.948	58	12	550	11.0	173, +13	3.973	110	09	37, 125
*		**10**	**14.3**†						**7.7**†	**172, +16**	**9.895**	**85**	**05**	**36, 125** (3°, 5°)‡

NOTE: N_{sp} = number of specimens; N_{SA} = number of samples; M_1 = intensity by 10^{-6} (e.m.u. system); R = resultant vector giving each sample unit weight; k precision estimate (Fisher 1953); α_{95} = radius; D is east of true north; I is positive downwards.

*Bold-faced characters give the value for the group using the site as unit weight; italicized site 7 has been excluded in these calculations.

†Logarithm mean of the site means.

‡Semi-axis (δp, δm) of the ellipse of confidence ($P = 0.05$)

§9 samples for M_N and 3 for M_T (see text).

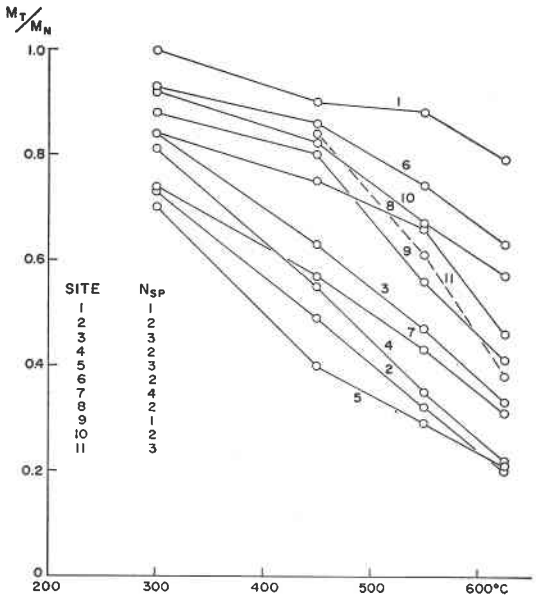


FIG. 2. Thermal decay of intensities of specimens from each site normalized to unity at room temperature. N_{sp} indicates the number of specimens contributing to each curve.

Results

One to four specimens from each site were heated to successively higher temperatures (up to 650 °C) and cooled in near zero field. The thermal decay curves (Fig. 2) show a wide range of blocking temperatures; except at site 7, the direction changes between successive heatings were usually <10°. The overall direction deviations between unheated and heated (at 650 °C) specimens were also small (<15°) except at site 9, where they were approximately 30°. This indicates that most of the natural remanent magnetization was acquired in a field of constant polarity and that no field reversal took place during its acquisition.

Negligible direction changes (except for site 7) occurred on thermal cleaning in the 550 to 650 °C range and therefore a 550 °C cleaning temperature was generally adopted. The sample directions before and after cleaning are shown in Fig. 3 and the site mean values and statistics given in Table I.

At site 7 (Fig. 4) six of the nine samples seemed to have random direction changes after each heating. This was observed at temperatures as low as 450 °C. However, from a comparison of the direction changes and the orientation of the specimens, these changes can be attributed to a small downward field in the oven (undercompensation of the Z component of the earth's field). Repeat measurements have shown that the field cancellation is better than 100γ and most of the time better than 50γ depending on diurnal variations. Therefore, certain sediments may acquire some magnetization in very low

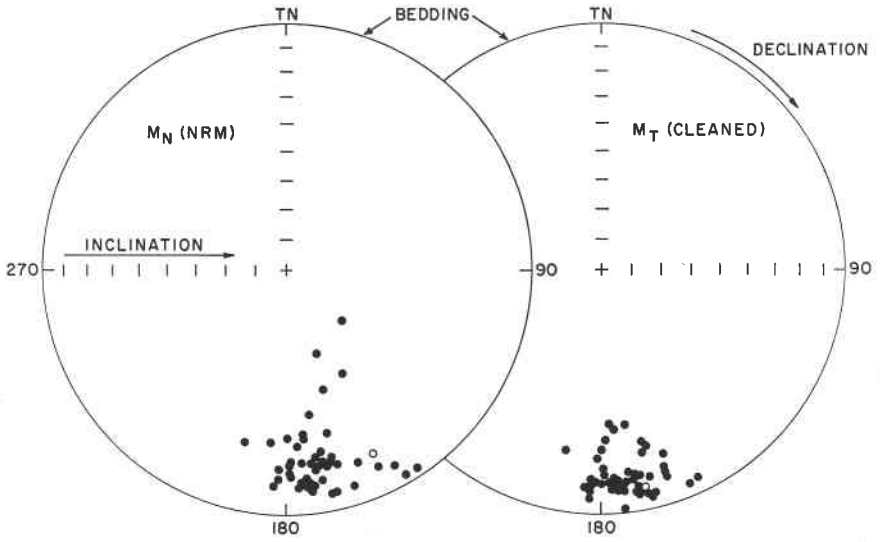


FIG. 3. Sample directions before (M_N) and after (M_T) cleaning at temperature T given in Table I. Equal-area polar projection. Sample directions from site 7 are given in Fig. 4.

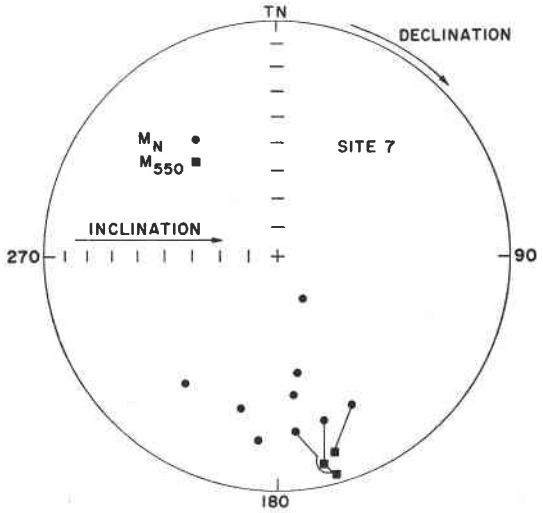


FIG. 4. Sample directions with respect to bedding before (M_N) and after (M_{550}) cleaning at 550 °C. Equal-area polar projection.

fields, and these observations confirm the importance of field cancellation in thermal experiments. Three samples did not show this effect and their direction mean is given in Table I; nevertheless, this site has not been used in the computation of the final statistics.

Fold Test

Site mean directions are plotted in Fig. 5. Without correction for dip, the relevant statistics are $N = 10$, $R = 9.26$, $k_1 = 12$. After correction, the precision is much improved: $R = 9.89$, $k_2 = 85$. The ratio k_2/k_1 is much larger than the value $F_{.005}(18,18) = 3.6$ indicating that, at the 0.005 level of significance, the magnetization remaining after cleaning was acquired before folding, that is before Pictou time.

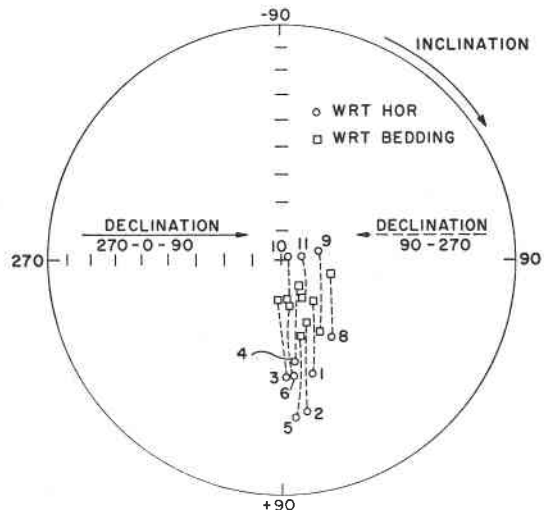


FIG. 5. Fold test after thermal cleaning. Equal-area equatorial projection.

An *F*-distribution test of between and within site dispersions (Watson 1956) gives a value of 4.7, which is greater than the tabled value $F_{.005}(18, 76) = 2.4$. Therefore, even at the 0.005 level of significance, the null hypothesis does not hold, indicating that the site means have a significant dispersion.

Secular Variation

The standard deviation of the site means is 9° and much smaller than the 18° calculated as the direction dispersion of the present field (Creer 1955) at that latitude. This difference could be real and indicative of smaller secular variation in late Poleozoic time than at present. Alternatively, diagenetic magnetization processes may have integrated the field at each site over considerable time spans: the effect of this would be to reduce the dispersion (Roy and Robertson 1968).

Summary of Upper Paleozoic Results from the Dominion Observatory

Ten pole positions are shown in Fig. 6 as a summary of work done at this laboratory. The results have been obtained with the same apparatus and under uniform experimental conditions and uniform statistical analysis. Information relating to the age of formation, age of magnetization, and time of folding (if any) is tabulated in Table II. The errors drawn in Fig. 6 are the elliptical standard errors derived from the standard errors of the site means for that formation. No ellipse has been drawn for the Perry Formation; that pole position is only a very approximate result (Robertson *et al.* 1968).

The poles show a systematic trend with time. Although the ellipses of confidence of two adjacent formations (of Table II) often overlap, the pole displacement never regresses, that is, the younger the formation, the more eastward the pole. The magnitude of the overall displacement (27°) is in good agreement with

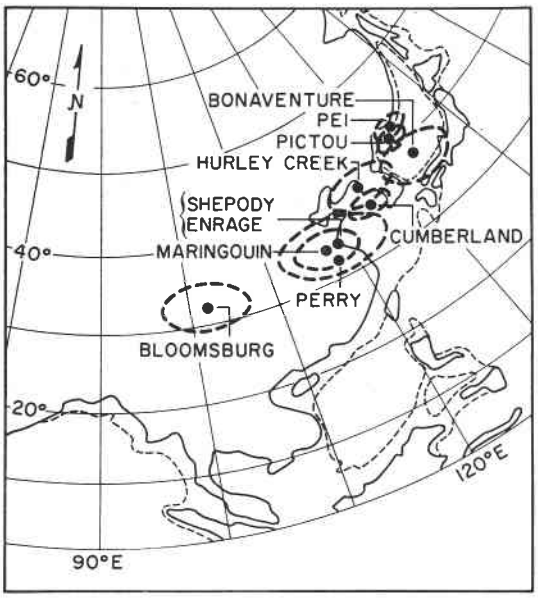


FIG. 6. Paleozoic pole positions relative to eastern North America. The elliptical standard error has been derived from the standard error of the site means at the sampling localities.

Paleozoic European data obtained from demagnetization studies (Storetvedt 1968), which show a difference of about 30° between pole positions derived either from lower Paleozoic or from upper Paleozoic rock formations. Similar patterns observed in two continents indicate that a polar change in the upper Paleozoic rocks, perhaps as large as 25° to 30°, is conceivable. An apparent polar change, at approximately the same time, although of greater magnitude, was also observed in Australia (Irving 1966) and Africa (McElhinny and Opdyke 1968).

Carboniferous Geomagnetic Field

Irving (1966) has suggested that a polar change took place in mid-Carboniferous time. Storetvedt (1968) also placed most of the change observed in the Paleozoic pole relative to Europe within the Carboniferous Period. The results presented here are pertinent to this question, since seven of the nine pole positions have been obtained from upper or mid-Carboniferous rock formations. The range of the apparent polar change given by these formations is 16° (Maringouin Formation to Prince Edward Island sediments). Various sources of error, such as secular variation or inclination error, which could possibly produce divergence of this magnitude, were discussed in the articles cited and were not considered to be appreciable. For example, there was no evidence in the sediments of the Hopewell Group (Enragé, Shepody, and Maringouin Formations) of an inclination error, and in most of the other formations the inclination was so shallow that an inclination error would have been negligible. Secular variation was taken into account in all papers and it was concluded that its effect on the mean direction was not appreciable. It then seems reasonable to take 16° as a reasonable

TABLE II

Age information relating to certain Paleozoic rock formations

Formation or group	Probable age of formation	Age of magnetization		Polarity	Reference
		Prior to or	Determined by		
P.E.I. sediments	Early Permian or late Pennsylvanian	Permian	Divergence from Triassic or later directions	+	Roy (1966)
Pictou Gp. (Tormentine Fm.)	Late Pennsylvanian	Permian	"	+	Roy (1966)
Bonaventure Fm.	Pennsylvanian (?)	Permian	"	+	Roy (1966)
Hurley Creek Fm.	Late Pennsylvanian	Permian	"	+	Roy *et al.* (1968)
Cumberland Gp.	Middle Pennsylvanian	Pictou	fold test	+	this paper
Enragé Fm.	Early Pennsylvanian	Cumberland Gp.	fold test	+	Roy and Park (1969)
Shepody Fm.	Late Mississippian	Cumberland Gp.	fold test	M*	Roy and Park (1969)
Maringouin Fm.	Late Mississippian	Cumberland Gp.	fold test	+	Roy and Robertson (1968)
Perry Fm.	Late Devonian	Early Pennsylvanian	folding effect	+	Robertson *et al.* (1968)
Bloomsburg Fm.	Late Silurian	Appalachian folding	fold test	−(M?)*	Roy *et al.* (1967)

*Mixed polarity.

estimate of the apparent polar change in eastern North America between late Mississippian and Permian times. It is possible that some of this change took place in early Pennsylvanian time. The magnetization of the Hopewell Group was acquired before a folding, which according to Gussow (1953) occurred during Cumberland time. This suggests different times of magnetization for the Cumberland and Hopewell Groups. The difference between the poles of the Cumberland and Hopewell strata (Maringouin and Shepody–Enragé of Fig. 6) may therefore be indicative of an apparent polar shift at that time.

The magnetizations of the Cumberland Group and younger formations were all southerly directed, indicating that the long period of positive polarity which has been associated with the Permian and late Carboniferous (Kiaman) Periods may have started in Eastern Canada as early as Cumberland time. It is of interest to note that the exact age of the Bonaventure Formation is uncertain. The Bonaventure has been associated with the Cannes de Roche Formation, which carries plant remains "which may indicate a late Mississippian but more probably a Pennsylvanian age" (Weeks 1963, p. 171). The paleomagnetic results (see Fig. 6) obtained favor a Pennsylvanian rather than a Mississippian age.

There are not enough data from the lower Paleozoic to permit a conclusion on the field at that time. The Bloomsburg pole position is the only one obtained from demagnetization studies of North American formations. The magnetization remaining after cleaning may or may not be Silurian. It has been shown to be pre-Appalachian orogeny and is probably pre-Cumberland, since some of the directions are northerly directed. Furthermore, divergence of its pole position from that of the Hopewell Group suggests that it is pre-Pennsylvanian. However, it is evident that before the direction and movement of the lower Paleozoic field in North America can be determined, many more data must be gathered.

If the earth's field during later Paleozoic times was axial, these data indicate that the region comprising the Maritimes and the Gulf of St. Lawrence was then near the equator; middle Carboniferous rock formations (Hopewell Group) give a paleo-latitude for that region of 07° to 12° S and the younger rock formations (Kiaman) a paleolatitude of 02° N to 03° S. Therefore, if upper Paleozoic oil fields are likely to be found in low paleogeographic latitudes (Irving and Gaskell 1962), then one of the criteria for finding oil in the Maritimes and Gulf of St. Lawrence region is fulfilled.

Acknowledgments

I wish to thank Dr. W. A. Robertson of the Dominion Observatory for his help in field collecting and Dr. D. G. Kelly of the Geological Survey of Canada for enlightening discussions on the geology of the Maritime provinces. I am very grateful to the Geophysics Division of the Geological Survey of Canada for putting at my disposition their electric oven. The assistance of Mr. J. K. Park with the measurements was greatly appreciated.

BELL, W. A. 1944. Carboniferous rocks and fossil floras of Northern Nova Scotia. Geol. Surv. Can., Mem. 238, pp. 29–30.

CANADA, DEPARTMENT OF MINES AND RESOURCES. 1938. Oxford sheet. Geol. Surv. Can., Map 410A.

CREER, K. M. 1955. A preliminary paleomagnetic study of certain rocks in England and Wales. Unpublished Ph.D. thesis, Univ. Cambridge, Cambridge, England.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. Roy. Soc. (London) Ser. A, **217**, pp. 295–305.

GUSSOW, W. C. 1953. Carboniferous stratigraphic and structural geology of New Brunswick, Canada. Bull. Amer. Assoc. Petrol. Geol., **37**, pp. 1713–1816.

IRVING, E. 1966. Paleomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events. J. Geophys. Res., **71**, pp. 6025–6051.

IRVING, E. and GASKELL, T. F. 1962. The paleogeographic latitude of oil fields. Geophys. J., **7**, pp. 54–63.

MCELHINNY, M. W. and OPDYKE, N. D. 1968. The paleomagnetism of some Carboniferous glacial varves from Central Africa. J. Geophys. Res., **73**, pp. 689–696.

ROBERTSON, W. A., ROY, J. L., and PARK, J. K. 1968. Magnetization of the Perry formation of New Brunswick, and the rotation of Newfoundland. Can. J. Earth Sci., **5**, pp. 1175–1181.

ROY, J. L. 1966. Désaimantation thermique et analyse statistique des directions de sédiments carbonifères de l'est du Canada. Can. J. Earth Sci., **3**, pp. 139–161.

ROY, J. L., OPDYKE, N. D., and IRVING, E. 1967. Further paleomagnetic results from the Bloomsburg formation. J. Geophys. Res., **72**, pp. 5075–5086.

ROY, J. L. and PARK, J. K. 1969. Paleomagnetism of the Hopewell Group, New Brunswick. J. Geophys. Res., **74**, pp. 594–604.

ROY, J. L. and ROBERTSON, W. A. 1968. Evidence for diagenetic remanent magnetization in the Maringouin Formation. Can. J. Earth Sci., **5**, pp. 275–285.

ROY, J. L., ROBERTSON, W. A., and PARK, J. K. 1968. Stability of the magnetization of the Hurley Creek formation. J. Geophys. Res., **73**, pp. 697–702.

STORETVEDT, K. M. 1968. A synthesis of Paleozoic data for Europe. Earth Planet. Sci. Lett., **3**, pp. 444–448.

WATSON, G. S. 1956. Analysis of dispersion on a sphere. Monthly Notices Roy. Astron. Soc., Geophys. Supp., **7**, pp. 153–159.

WEEKS, L. J., 1963. The Appalachian region. *In* Geology and economic minerals of Canada (4th Edn. 1957). Geol. Surv. Can., Econ. Geol. Ser. No. 1, pp. 123–205.